Cost Economics of Vector Databases (Part 1: The Managed Trap)

Fully managed vector databases like Pinecone (pod-based), Milvus managed, and AWS OpenSearch rely on always
For an enterprise index with 1,000,000 vectors of 768 dimensions (approx. 3.1 GB raw float32 embeddings),
traditional HNSW graphs require holding both vectors and graph edges in memory 24/7.
On Pinecone, 1M 768d vectors require at least two s1 or p1 pods for redundancy and performance.
At approximately $70 per pod-month, an idle or lightly queried index costs $140 to $280 per month.
At 10,000,000 vectors, memory requirements scale to 32 GB - 64 GB of RAM, costing $1,400 to $2,800+ monthly.

This pricing model creates an economic mismatch for read-light workloads (e.g. internal knowledge bases with 500 c
Organizations pay primarily for idle RAM allocations rather than actual search compute.

Cost Economics of Vector Databases (Part 2: Embedded & Serverless)

Serverless and embedded vector engines like LanceDB, DuckDB-vss, and sqlite-vec decouple compute from storag
LanceDB stores compressed IVF-PQ vectors on block storage (local NVMe SSD or cloud object stores like AWS S3
At 100K vectors (300 MB storage): S3 storage costs roughly $0.01/month, and local disk cost is effectively $0.00.
At 1M vectors (3 GB storage): S3 storage costs roughly $0.07/month.
At 10M vectors (30 GB storage): S3 storage costs roughly $0.70/month.

Compute costs in an embedded model are co-located within the application process or serverless container (e.g. AW
During idle periods, compute cost is zero. Under query load, CPU cycles are consumed strictly per request.
The accepted engineering trade-off for this 100x cost reduction is p95 retrieval latency: LanceDB achieves 5-15ms c
compared to 2-5ms for pure in-memory HNSW, but saves thousands of dollars per year.